

Cilostazol Attenuates Hepatic Steatosis and Intestinal Disorders in Nonalcoholic Fatty Liver Disease.

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Nonalcoholic fatty liver disease (NAFLD) is one of the most common chronic liver diseases in the world, which begins with liver lipid accumulation and is associated with metabolic syndrome. Also, the name chosen to replace NAFLD was metabolic dysfunction-associated steatotic liver disease (MASLD). We performed focused drug screening and found that Cilostazol effectively ameliorated hepatic steatosis and might offer potential for NAFLD treatment. Our aim was to investigate the therapeutic effects of Cilostazol on the glycolipid metabolism and intestinal flora in NAFLD mice and explore the specific mechanism. In this study, 7-week-old male C57BL/6J mice were fed a high-fat diet (HFD) for 8 weeks to induce NAFLD, and then treated with intragastric administration for 12 weeks. The results showed that Cilostazol inhibited liver lipid de novo synthesis by regulating the AMPK-ACC1/SCD1 pathway and inhibited liver gluconeogenesis by the AMPK-PGC1 α -G6P/PEPCK pathway. Cilostazol improved the intestinal flora diversity and intestinal microbial composition in the NAFLD mice, and specifically regulated *Desulfovibrio* and *Akkermansia*. In addition, Cilostazol increased the level of short-chain fatty acids in the NAFLD mice to a level similar to that in the blank Control group. Cilostazol reduces liver lipid accumulation in NAFLD mice by improving glucose and lipid metabolism disorders and intestinal dysfunction, thereby achieving the purpose of treating NAFLD.

Metabolic dysfunction-associated fatty liver disease and implications for cardiovascular risk and disease prevention.

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The newly proposed term "metabolic dysfunction-associated fatty liver disease" (MAFLD) is replacing the old term "non-alcoholic fatty liver disease" (NAFLD) in many global regions, because it better reflects the pathophysiology and cardiometabolic implications of this common liver disease. The proposed change in terminology from NAFLD to MAFLD is not simply a single-letter change in an acronym, since MAFLD is defined by a set of specific and positive diagnostic criteria. In particular, the MAFLD definition specifically incorporates within the classification recognized cardiovascular risk factors. Although convincing evidence supports a significant association between both NAFLD and MAFLD, with increased risk of CVD morbidity and mortality, neither NAFLD nor MAFLD have received sufficient attention from the Cardiology community. In fact, there is a paucity of scientific guidelines focusing on this common and burdensome liver disease from cardiovascular professional societies. This Perspective article discusses the rationale and clinical relevance for Cardiologists of the newly proposed MAFLD definition.

The Gut Microbiome and Ferroptosis in MAFLD.

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Metabolic-associated fatty liver disease (MAFLD) is a new disease definition, and is proposed to replace the previous name, nonalcoholic fatty liver disease (NAFLD). Globally, MAFLD/NAFLD is the most common liver disease, with an incidence rate ranging from 6% to 35% in adult populations. The pathogenesis of MAFLD/NAFLD is closely related to insulin resistance (IR), and the genetic susceptibility to acquired

metabolic stress-associated liver injury. Similarly, the gut microbiota in MAFLD/NAFLD is being reevaluated by scientists, as the gut and liver influence each other via the gut-liver axis. Ferroptosis is a novel form of programmed cell death caused by iron-dependent lipid peroxidation. Emerging evidence suggests that ferroptosis has a key role in the pathological progression of MAFLD/NAFLD, and inhibition of ferroptosis may become a novel therapeutic strategy for the treatment of NAFLD. This review focuses on the main mechanisms behind the promotion of MAFLD/NAFLD occurrence and development by the intestinal microbiota and ferroptosis. It outlines new strategies to target the intestinal microbiota and ferroptosis to facilitate future MAFLD/NAFLD therapies.

Metabolic dysfunction-associated steatotic liver disease-related hepatic fibrosis increases risk of insulin resistance, type 2 diabetes, and chronic kidney disease.

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Metabolic dysfunction-associated steatotic liver disease (MASLD) (previously called nonalcoholic fatty liver disease, NAFLD) is associated with cardiometabolic risk factors and chronic kidney disease (CKD). However, evidence is lacking regarding whether the severity of fibrosis is affected by these risk factors and diseases and to what degree. We aimed to determine the correlation between these factors and vibration-controlled transient elastography-determined liver stiffness measurements (LSMs) and controlled attenuation parameter (CAP) values in a sample of the US population. Data from the 2017-2018 cycle of the National Health and Nutrition Examination Survey were pooled. The association between LSM and cardiometabolic risk factors and CKD was assessed using generalized linear or logistic regression analyses. In multivariate regression analyses, CAP and BMI were adjusted as confounders. Of 3647 participants, 2079 (57.1%) had NAFLD/MASLD [weighted prevalence 54.8%; 95% confidence interval (CI) 51.8-57.9%]; the weighted prevalence of significant fibrosis (LSM $\geq$ 7.9 kPa) was 9.7% (95% CI 8.2-11.3%). Log LSM was associated with higher levels of homeostatic model assessment of insulin resistance ($\beta = 2.19$; $P = 0.017$), hepatic steatosis (CAP > 248 dB/m) [odds ratio (OR) 3.66; 95% CI 2.22-6.02], type 2 diabetes (OR 2.69; 95% CI 1.72-4.20), and CKD (OR 1.70; 95% CI 1.24-2.34). These correlations did not change notably after adjustments were made for waist circumference, CAP, and BMI. LSM and CAP, although influenced by waist circumference and BMI, are good indicators of hepatic fibrosis and steatosis. LSM is associated with insulin resistance, diabetes, and CKD independent of hepatic steatosis and obesity.

Prognostic value of metabolic dysfunction-associated steatotic liver disease over coronary computed tomography angiography findings: comparison with no-alcoholic fatty liver disease.

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Metabolic dysfunction-associated steatotic liver disease (MASLD) is the proposed name change for non-alcoholic fatty liver disease (NAFLD). This study aimed to investigate the association of cardiovascular disease risk with MASLD and NAFLD in patients who underwent clinically indicated coronary computed tomography angiography (CCTA). This retrospective study included 2289 patients (60% men; mean age: 68 years) with no history of coronary artery disease who underwent CCTA. The steatotic liver was defined as a hepatic-to-spleen attenuation ratio of < 1.0 on CT just before CCTA.

MASLD is defined as the presence of hepatic steatosis along with at least one of the five cardiometabolic risk factors. Adverse CCTA findings were defined as obstructive and/or high-risk plaques. Major adverse cardiac events (MACE) encompassed composite coronary events, including cardiovascular death, acute coronary syndrome, and late coronary revascularization. MASLD and NAFLD were identified in 415 (18%) and 368 (16%) patients, respectively. Adverse CCTA findings were observed in 40% and 38% of the patients with MASLD and with NAFLD, respectively. Adverse CCTA findings were significantly associated with MASLD ($p = 0.007$) but not NAFLD ($p = 0.253$). During a median follow-up of 4.4 years, 102 (4.4%) MACE were observed. MASLD was significantly associated with MACE (hazard ratio 1.82, 95% CI 1.18-2.83, $p = 0.007$), while its association with NAFLD was not significant ($p = 0.070$). By incorporating MASLD into a prediction model of MACE, including the risk score and adverse CCTA findings, global chi-squared values significantly increased from 87.0 to 94.1 ($p = 0.008$). Patients with MASLD are likely to have a higher risk of cardiovascular disease than those with NAFLD. Concurrent assessment of MASLD during CCTA improves the identification of patients at a higher risk of cardiovascular disease among those with clinically indicated CCTA.

Management strategies for metabolic dysfunction-associated steatotic liver disease (MASLD).

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Metabolic dysfunction-associated steatotic liver disease (MASLD) is characterized by hepatic steatosis that is confirmed by imaging or histology in the setting of at least 1 metabolic risk factor in the absence of significant alcohol consumption. Nonalcoholic steatohepatitis, or NASH, was recently renamed metabolic dysfunction-associated steatohepatitis (MASH); it represents the progressive form of MASLD. MASH is defined by hepatic steatosis, lobular inflammation, and ballooning degeneration (hepatocellular injury) in a characteristic histologic pattern. Multiple pathophysiologic mechanisms underlie the development of MASLD, and multiple factors (eg, metabolic, hormonal, genetic, nutritional, and epigenetic components) are related to liver injury. MASH has a prevalence in the United States of 1% to 6%, and it is expected to rise in the next decade. Individuals living with MASH frequently suffer from comorbidities such as type 2 diabetes and cardiovascular disease. Several guidelines have been published to support the timely diagnosis of MASH that incorporate noninvasive tests that obviate the need for liver biopsy. Multiple MASH treatment options are in various stages of development. The THR- β agonist resmetirom, approved by FDA in March 2024, offers a liver-directed treatment for those patients living with moderate to severe fibrosis without cirrhosis. Considering the progressive nature of the disease and the availability of a treatment that can be initiated early to halt MASH progression, patients who have risk factors for MASH should urgently be encouraged to visit their health care providers for MASH screening.

Metabolic Muddle. MASLD and MASH on the Horizon.

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NAFLD (non-alcoholic fatty liver disease) and NASH (non-alcoholic steatohepatitis) are time-honored acronyms, with widely popular acceptance. Experts now recommend discarding them in favor of MASLD for "metabolic dysfunction-associated steatotic liver disease" and MASH for "metabolic

dysfunction-associated steatohepatitis." The reasons for this change are explored and an argument about why the change is confusing, is advanced. Should these acronyms become clinically popular, risk assessment manuals will require updates.

Evaluation of Google Search Trends for Liver Diseases in Europe.

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Chronic liver diseases belong to the most common diseases worldwide and are associated with increased morbidity and mortality. Although more than one in three adults are estimated to have metabolic dysfunction-associated steatotic liver disease (MASLD), awareness of this condition is low amongst the general public, health care professionals and policy makers. However, meaningful knowledge transfer is essential for raising awareness and improving prevention and treatment. This study set out to investigate the use of the major internet search engine to understand how knowledge transfer has evolved by analyzing liver-related searches trends. We investigated Google search trends by measuring the number of hits relating to liver diseases between 2004 and 2021 in seven languages and European countries but also worldwide. All analyses were performed in R using the R Google trends package gtrendsR. We found that interest in MASLD [formerly non-alcoholic fatty liver disease (NAFLD)] has generally increased over time, but that interest in metabolic associated steatohepatitis (MASH) - the most severe form of MASLD - has decreased. Interest in viral hepatitis C has decreased, whereas the number of queries regarding viral hepatitis B have been stable but dominated by interest in vaccination for it. Recent medical developments (in viral hepatitis) did not lead to a noticeable change in overall search behavior. Users preferred searching using their native language and less complex medical terms and acronyms (e.g., fatty liver instead of NAFLD). In the last two decades, Google search trends have followed the general development in the field of hepatology. Searches were dominated by non-experts and are not being rapidly influenced by novel scientific developments. Also, users preferred search terms in their native languages rather than English and tended to avoid complex medical search terms. Awareness and communication strategies around MASLD should consider these preferences when addressing the general public.

A multisociety Delphi consensus statement on new fatty liver disease nomenclature.

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The principal limitations of the terms NAFLD and NASH are the reliance on exclusionary confounder terms and the use of potentially stigmatising language. This study set out to determine if content experts and patient advocates were in favour of a change in nomenclature and/or definition. A modified Delphi process was led by three large pan-national liver associations. The consensus was defined a priori as a supermajority (67%) vote. An independent committee of experts external to the nomenclature process made the final recommendation on the acronym and its diagnostic criteria. A total of 236 panellists from 56 countries participated in 4 online surveys and 2 hybrid meetings. Response rates across the 4 survey rounds were 87%, 83%, 83%, and 78%, respectively. Seventy-four percent of respondents felt that the current nomenclature was sufficiently flawed to consider a name change. The terms "nonalcoholic" and "fatty" were felt to be stigmatising by 61% and 66% of respondents, respectively. Steatotic liver disease was chosen as an overarching term to encompass the various aetiologies of steatosis. The term steatohepatitis was felt to be an important pathophysiological concept that should be retained. The name

chosen to replace NAFLD was metabolic dysfunction-associated steatotic liver disease (MASLD). There was consensus to change the definition to include the presence of at least 1 of 5 cardiometabolic risk factors. Those with no metabolic parameters and no known cause were deemed to have cryptogenic steatotic liver disease. A new category, outside pure metabolic dysfunction-associated steatotic liver disease, termed metabolic and alcohol related/associated liver disease (MetALD), was selected to describe those with metabolic dysfunction-associated steatotic liver disease, who consume greater amounts of alcohol per week (140-350 g/wk and 210-420 g/wk for females and males, respectively). The new nomenclature and diagnostic criteria are widely supported and non-stigmatising, and can improve awareness and patient identification.

Clinical profiles and mortality rates are similar for metabolic dysfunction-associated steatotic liver disease and non-alcoholic fatty liver disease.

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Recently, the term metabolic dysfunction-associated steatotic liver disease (MASLD) has replaced non-alcoholic fatty liver disease (NAFLD). Concern remains regarding whether the evidence generated under the NAFLD definition can be used for MASLD. We compared the clinical profile and outcomes of NAFLD to MASLD using tertiary care- and population-based data. Comparison data were obtained from our NAFLD database and the National Health and Nutrition Examination Survey (NHANES III). Clinical profiles and non-invasive tests (enhanced liver fibrosis [ELF] score, fibrosis-4 index [FIB-4] and vibration-controlled transient elastography) were compared. Mortality data were obtained from NHANES-National Death Index. All-cause mortality was assessed by Cox proportional hazards regression models and cause-specific mortality by competing risk analysis. There were 6,429 patients in the NAFLD database (age: 54 ± 12 years, 42% male, BMI 35.4 ± 8.3 , waist circumference 112 ± 17 cm, 52% type 2 diabetes). Average scores for ELF, FIB-4 and liver stiffness were 9.6 ± 1.2 , 1.69 ± 1.24 , 14.0 ± 11.8 kPa, respectively; 99% met MASLD criteria; 95% met MASLD on BMI only. Predictive accuracy of ELF and FIB-4 were identical between MASLD and NAFLD. We included 12,519 eligible participants from NHANES (age 43.00 years, 47.38% male, 22.70% obese, 7.28% type 2 diabetes, 82.51% ≥ 1 cardiometabolic criteria). Among the NHANES study population, there was excellent concordance between MASLD and NAFLD diagnoses: Cohen's kappa coefficient: 0.968 (95% CI 0.962-0.973) with 5.29% of NAFLD cases not meeting MASLD criteria. After a median follow-up of 22.83 years, there were no mortality differences between MASLD and NAFLD diagnoses (p values ≥ 0.05). NAFLD and MASLD are similar except individuals with MASLD seem to be older with slightly higher mortality risk, likely owing to cardiometabolic risk factors. Clinical profiles and non-invasive test thresholds were also identical. These data provide evidence that NAFLD and MASLD terminologies can be used interchangeably. For the small proportion of patients with NAFLD who do not meet MASLD criteria, further consideration is needed. In June 2023, new terminology (MASLD) was adopted to replace the term NAFLD as a means to better describe what the liver disease is rather than what it is not, as well as to potentially reduce stigma. Given that MASLD requires at least one cardiometabolic risk factor, questions were raised as to whether this change in the definition would nullify the similarities between NAFLD and MASLD and require new evidence to be generated for MASLD. We used our NAFLD database and a US population-based database to show that the vast majority of patients with NAFLD fulfill criteria for MASLD. Non-invasive tests performed similarly in both groups. Mortality risk was slightly higher in those with MASLD, which is attributed to the presence of cardiometabolic risks. These results provide evidence that data generated in the past three decades for NAFLD can be used interchangeably for MASLD.
